M. NABIL KHAIRI IKHSAN

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Summary

Applied Bachelor (D4) student in Electronics Engineering with experience in embedded systems development, industrial automation, and electronic system integration through academic and industrial projects. Competent in PCB design, PLC programming, Mini PC system integration, and troubleshooting. Oriented toward developing efficient embedded solutions tailored to industrial control and automation needs.

Core Competencies

- PCB Design & Layout
- Embedded Programming
- PLC Programming
- Control Systems Design
- Sensor Interfacing
- Thermal Analysis
- Electronic Measurements
- System Troubleshooting
- System Integration
- Reverse Engineering
- Power Distribution Analysis
- Technical Documentation

Experience

PT Aneka Tambang Tbk (ANTAM) UBPE Logam Mulia

Intern, Electrical Maintenance — **Jul 2026 – Present**

- Developed a Raspberry Pi-based warehouse inventory system to digitalize inventory transactions, support safety stock monitoring, and improve material procurement planning (70% project completion).
- Digitized Preventive Maintenance work instructions into visual documentation for 67 of 107 industrial machines (63%), improving maintenance standardization and technician comprehension.
- Developed KPI Control Boards for Pillar 1 of the Maintenance Maturity Level Assessment (MMLA) by standardizing performance indicators across Bureau, Work Unit, and Cost Center levels.

AMX UAV Technologies, Yogyakarta

Intern, Reverse Engineering (RE) Electric Drone Sprayer — **Feb 2026 – May 2026**

- Assemble and validate a UAV electrical system integrating 10+ electronic modules, including power distribution.
- Performed reverse engineering of the wiring diagram and Power Distribution Board (PDB) to support power distribution across 7+ UAV subsystems.
- Redesigned 2-layer PCB using EasyEDA, including high-current routing, thermal analysis, and component layout optimization to support fabrication readiness.
- Conducted power distribution testing, troubleshooting, and PCB design verification to ensure reliable electrical and mechanical integration.

Teaching Assistant at Yogyakarta State University (UNY)

Part-time, Instrumentation and Measurement Course — **Aug 2025 - Dec 2025**

- Mentored 14 undergraduate students in instrumentation and measurement laboratory sessions, covering the operation of multimeters, oscilloscopes, and other electronic test equipment.
- Evaluated 200+ laboratory reports throughout the semester and provided technical feedback to improve measurement accuracy and experimental analysis.
- Managed laboratory sessions in compliance with laboratory SOPs, maintaining zero equipment damage and workplace safety incidents throughout the teaching period.

PT PLN Icon Plus (Persero), Makassar

Intern, Field Network Technician — **Mar 2022 - Jun 2022**

- Conducted FTTH field surveys and network inspections with a target of approximately 30 Optical Distribution Points (ODPs) per day.
- Updated records for 2,510 Optical Distribution Points (ODPs) using digital mapping systems to support accurate network documentation and planning.
- Prepared and installed Optical Distribution Point (ODP) hardware, including adapters and 1:8 passive splitters, to support new unit deployment.

Technical Projects

AI-Based Robot Manipulator for Automated Waste Classification Using Raspberry Pi

Associated with Universitas Negeri Yogyakarta — **Feb 2026 - Jun 2026**

- Developed an AI-based robot manipulator using Raspberry Pi running Ubuntu to integrate computer vision, robotic control, and embedded system operation.
- Designed the manipulator structure in Autodesk Fusion 360, performed 3D printing, and calculated voltage and current requirements to ensure reliable electrical system integration.
- Trained and evaluated an object detection model for organic and inorganic waste classification using Python, OpenCV, and YOLO, achieving approximately 97% mAP@0.50, 98% Precision, and 95% Recall during validation.
- Integrated the trained model into the robotic system and conducted real-time object detection and classification experiments to support automated waste sorting.

Computer Vision-Based Vehicle Detection and Speed Estimation System

Associated with Universitas Negeri Yogyakarta — **Nov 2025 - Dec 2025**

- Developed a computer vision-based vehicle detection and classification system using YOLO11, OpenCV, and ByteTrack to identify four vehicle classes and estimate vehicle speed.
- Built a custom dataset by collecting data and automatically annotating 2,188 vehicle instances for model training.
- Implemented Perspective Transformation and Exponential Moving Average to improve speed estimation stability and vehicle risk analysis.
- Achieved 91.3% Mean Average Precision (IoU = 0.50), 91.3% Precision, and 91.0% Recall during model validation.

Autonomous Line Follower Robot Design and Development Using PID Control
Associated with Universitas Negeri Yogyakarta **Oct 2025 - Dec 2025**

- Designed the electronic and mechanical systems, including a custom PCB, robot chassis in Fusion 360, and integration of 8 photodiode sensors, a multiplexer, motor driver, and power management modules.
- Implemented sensor calibration and tuned PD control parameters to improve robot responsiveness and motion stability.
- Achieved a fastest track completion time of 7.9 seconds with stable and consistent performance during testing.

ROS2-Based Rover Robot Development with LiDAR Integration Using Jetson Orin Nano
Associated with Universitas Negeri Yogyakarta **Oct 2025 - Dec 2025**

- Developed a ROS2-based rover robot by integrating Jetson Orin Nano, LiDAR, and a motor driver to support robot navigation.
- Configured the ROS2 workspace, SSH communication, and developed Python-based motor control for robot operation.
- Implemented 360° LiDAR data visualization using Foxglove to display environmental scanning results.
- Successfully integrated the hardware and software systems, enabling robot control and LiDAR data visualization through ROS2.

Industrial Automation Logic Programming Using OpenPLC Editor
Associated with Universitas Negeri Yogyakarta **May 2025 - May 2025**

- Developed industrial automation logic using OpenPLC Editor and Ladder Diagram (LD) programming for sequential process control.
- Implemented standard PLC functions, including Timer (TON), Counter (CTU), arithmetic operations, and comparison functions to manage process sequencing and production cycle counting.
- Designed safety interlocks to ensure safe operation sequencing and prevent process conflicts during system operation.

Attendance Management System Development Using OpenCV and MySQL
Associated with Universitas Negeri Yogyakarta **Nov 2024 - Dec 2024**

- Developed a desktop-based attendance application using Python, Tkinter, OpenCV, and MySQL.
- Designed the graphical user interface (GUI) and integrated camera and database modules for identity validation and attendance recording.
- Implemented image capture, student ID validation, and attendance data storage using MySQL.

Three-Phase Induction Motor Control System Using Magnetic Contactors
Associated with Universitas Negeri Yogyakarta **May 2024 - Jun 2024**

- Designed and implemented a three-phase induction motor control system for forward-reverse operation using magnetic contactors.
- Designed the power and control circuits while integrating MCBs, push buttons, overload relays, and indicator modules into the control panel.
- Performed system testing by measuring starting current, running current, and phase-to-phase voltage to verify safe and reliable motor operation.

Leadership Experience

Himpunan Mahasiswa Vokasi Elektro dan Elektronika (HMVE UNY) **Jan 2025 - Dec 2025**
Vice Head of Entrepreneurship Division

- Led the procurement, design, and distribution of official jackets for 38 executive members, achieving 100% on-time completion in less than one month.
- Managed budgeting and financial reporting across four divisional programs, including capital planning, cost control, and profit evaluation.
- Increased divisional profit by over 100% within one leadership term through merchandise sales and branding initiatives.

Education

Universitas Negeri Yogyakarta | *Bachelor of Applied Science in Electronics Engineering* **Sep 2023 - Present**

- **GPA** : 3.69/4.00

SMK Telkom Makassar | *Access Network Engineering* **May 2020 – May 2023**

- **GPA** : 90.5/100

Skill

- **Hard Skills:** PCB Design, Electronic Hardware Design, Electrical System Analysis, Hardware Testing & Troubleshooting, Artificial Intelligence (AI) Development, Industrial Automation (PLC), Surface Mount Device (SMD) Soldering, PID Control System Tuning.
- **Software :** Microsoft Office, Linux, Raspberry Pi OS, Autodesk Fusion 360, EasyEDA, Omron CX-Programmer, Autodesk EAGLE, Proteus, Arduino IDE, Cisco Packet Tracer, Factory I/O, Webots, Falstad Circuit Simulator, OpenPLC Editor, OpenPLC Runtime, Robot Operating System 2 (ROS2), Python, C++, Firebase, Docker.
- **Language:** Indonesian (Native), English (Intermediate)

Certification

- Awareness K3 Electricity - **BPVP Sidoarjo Kementerian Ketenagakerjaan RI** **Apr 2026**
- Certificate in Reading and Identifying Passive Electronic Components - **BPVP Ambon Kementerian Ketenagakerjaan RI** **Apr 2026**
- Occupational Health and Safety Certificate - **PKBM & LPK ZIONA** **Feb 2026**
- Certificate Using Mechanical and Electrical Measuring Instruments - **BPVP Banyuwangi Kementerian Ketenagakerjaan RI** **Feb 2026**
- Certificate for Using Tools and Materials for Cell Phone Repair and Maintenance - **BLKPP Daerah Istimewa Yogyakarta** **Feb 2026**